

Prompt to za mało

Jak projektować kontekst dla
LLM-ów i automatyzować jego
poprawianie





- **8+ lat doświadczenia w ML**

- **Ostatnie 3 lata**

- LLM-y dla sektora prawniczego
- analiza dokumentów
- bazy wiedzy

- **Wcześniej**

- predictive maintenance
- anomaly detection

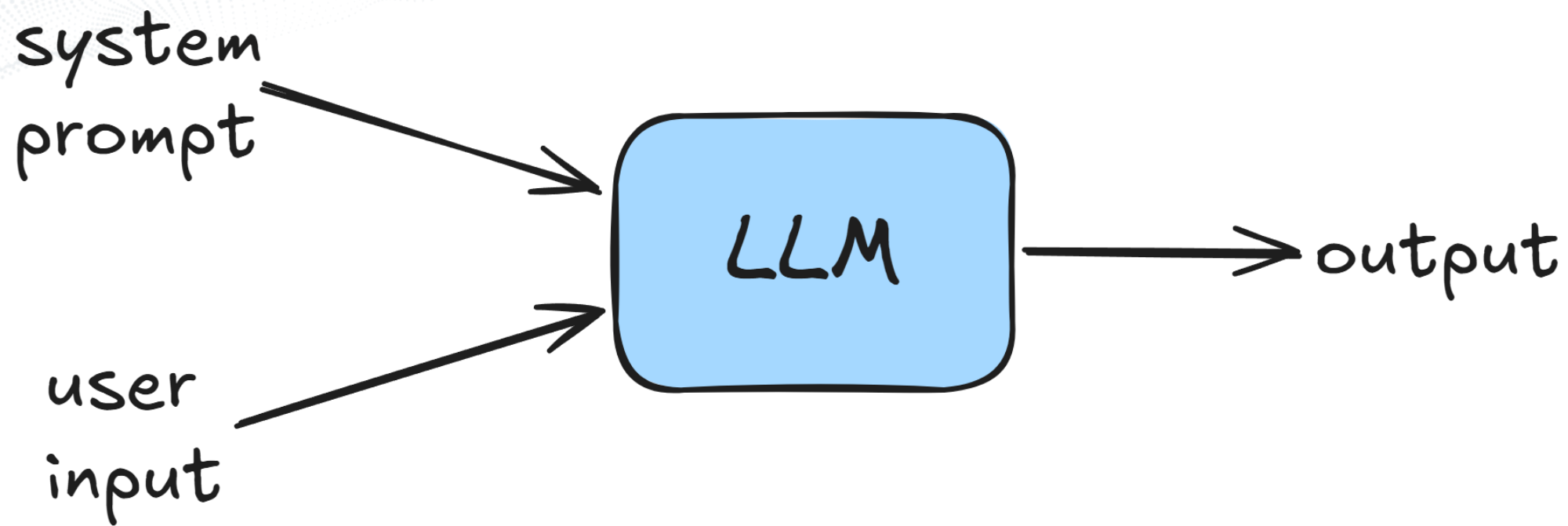


linkedin.com/in/hubertjegierski

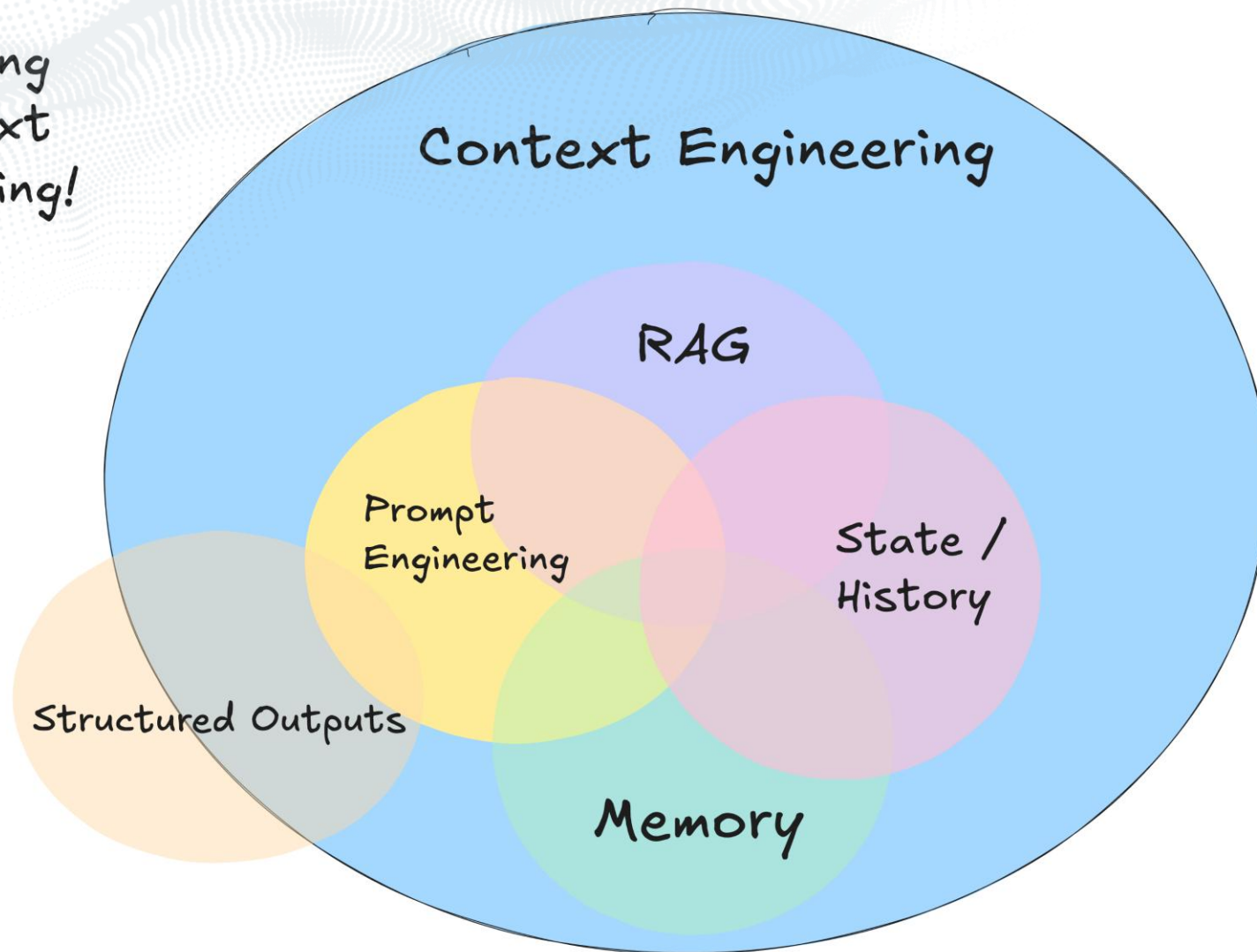


stoppedthinking.substack.com

Czym jest kontekst?



Everything
is Context
Engineering!

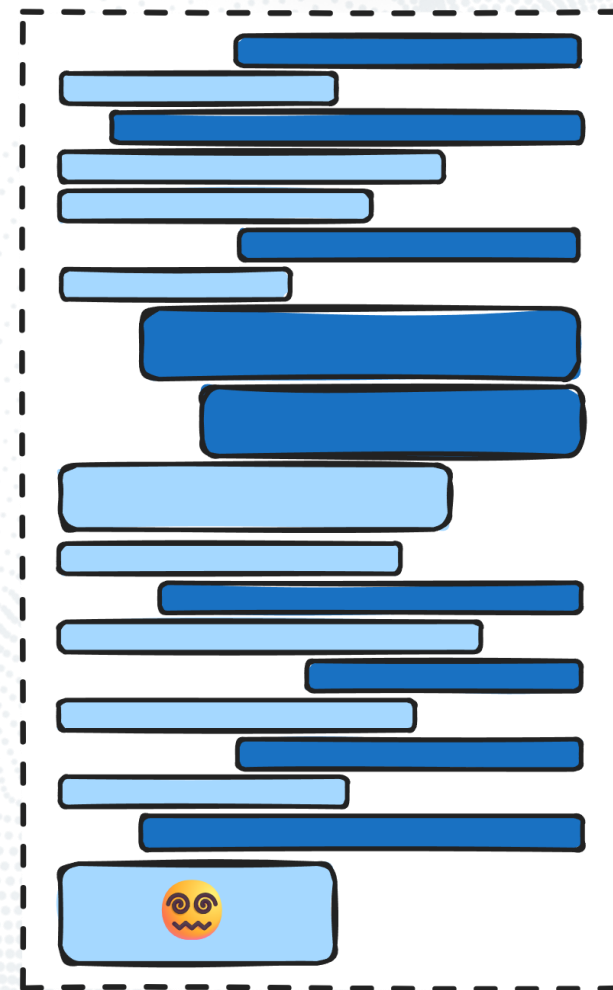




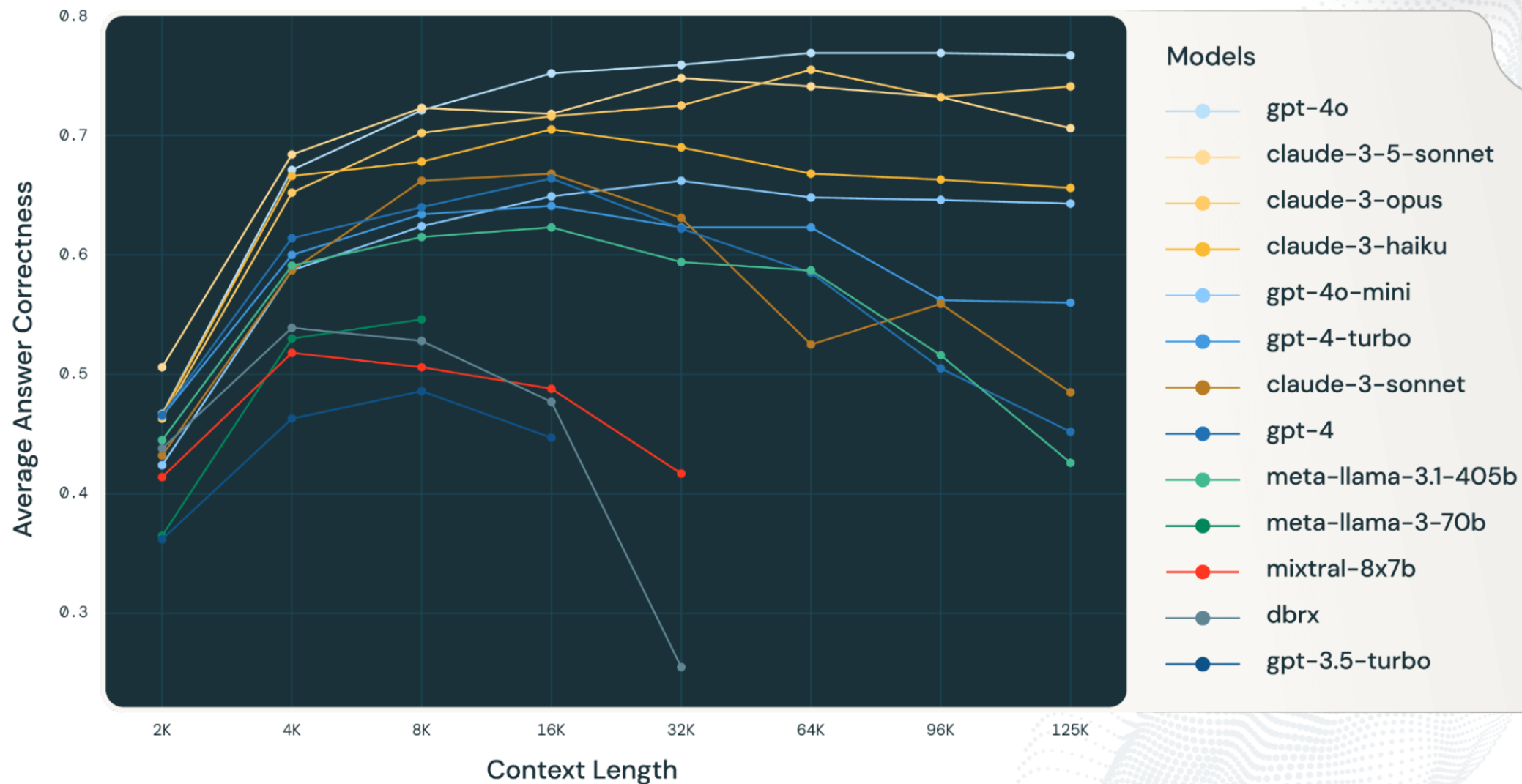
Dlaczego kontekst zawodzi?

1. Nadmiar słów

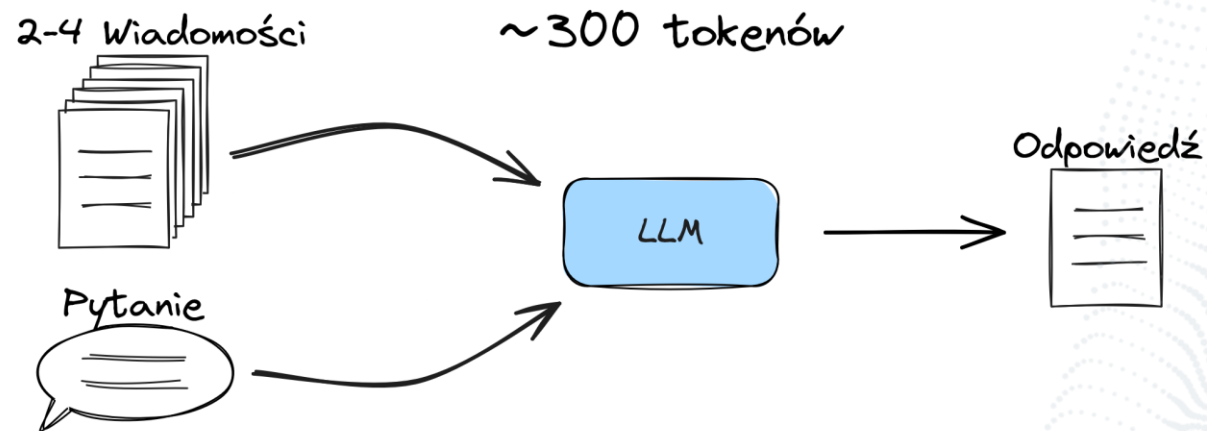
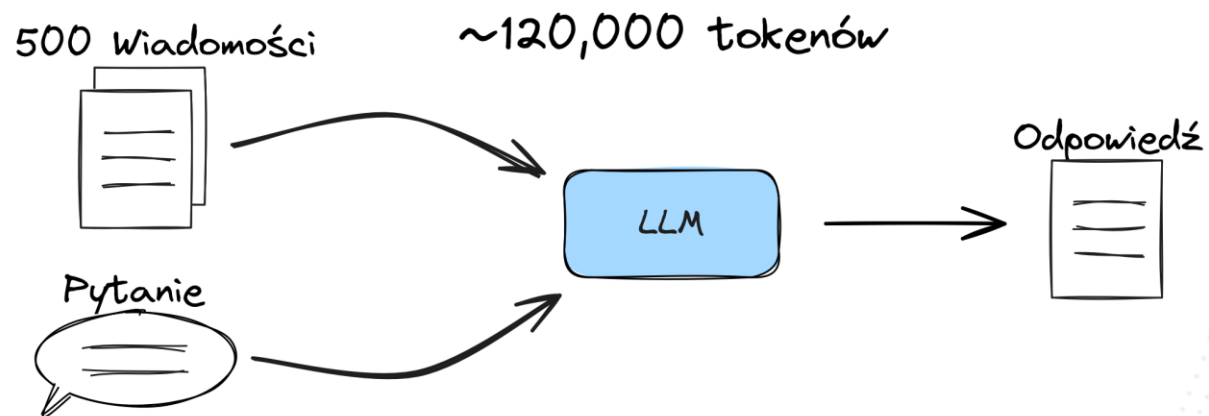
LLM-y wspierają gigantyczne konteksty, jednak nadmiar słów sprawia, że się w nim gubią. Model zapomina o swoim zadaniu, zaczyna powtarzać bezsensowne czynności albo zwraca absurdalne odpowiedzi.



Long Context



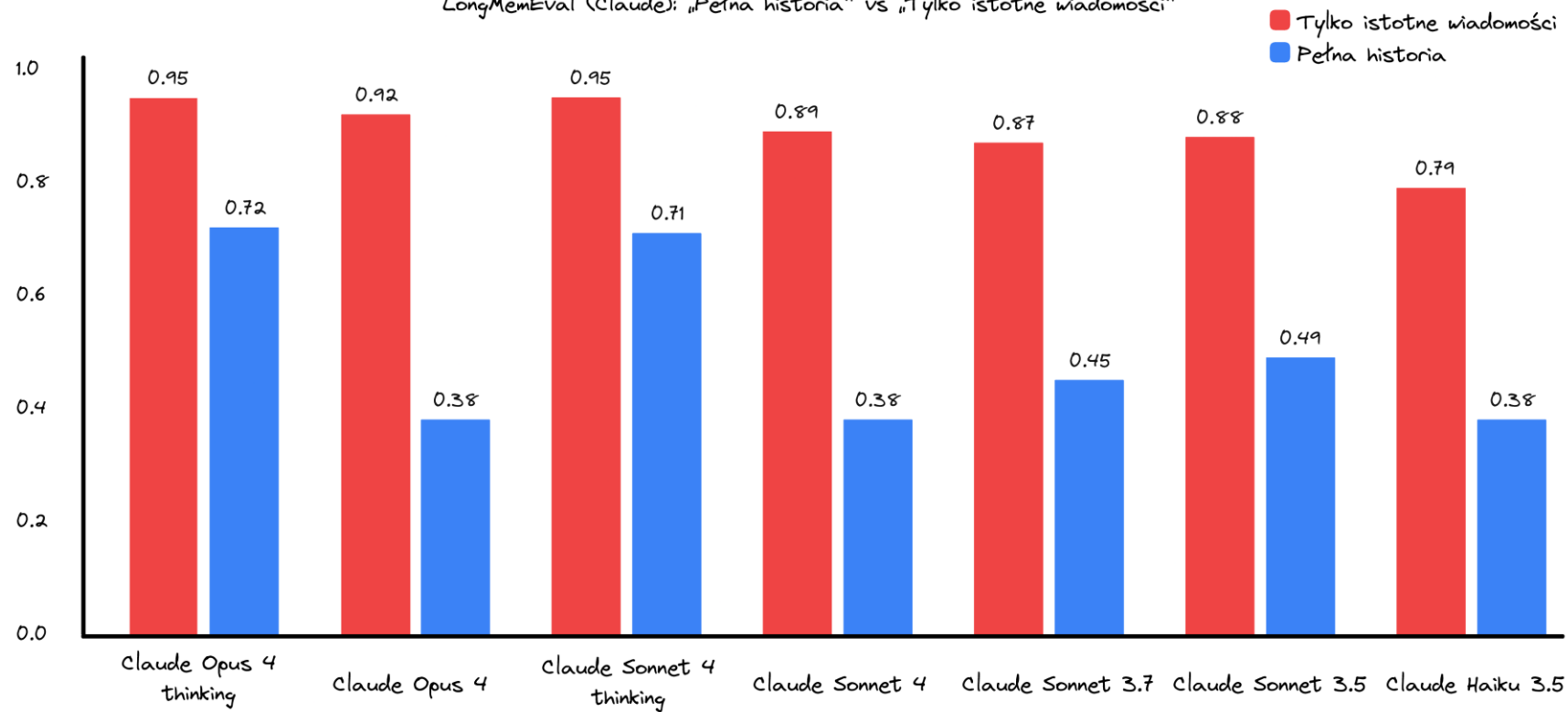
Long Context



Long Context

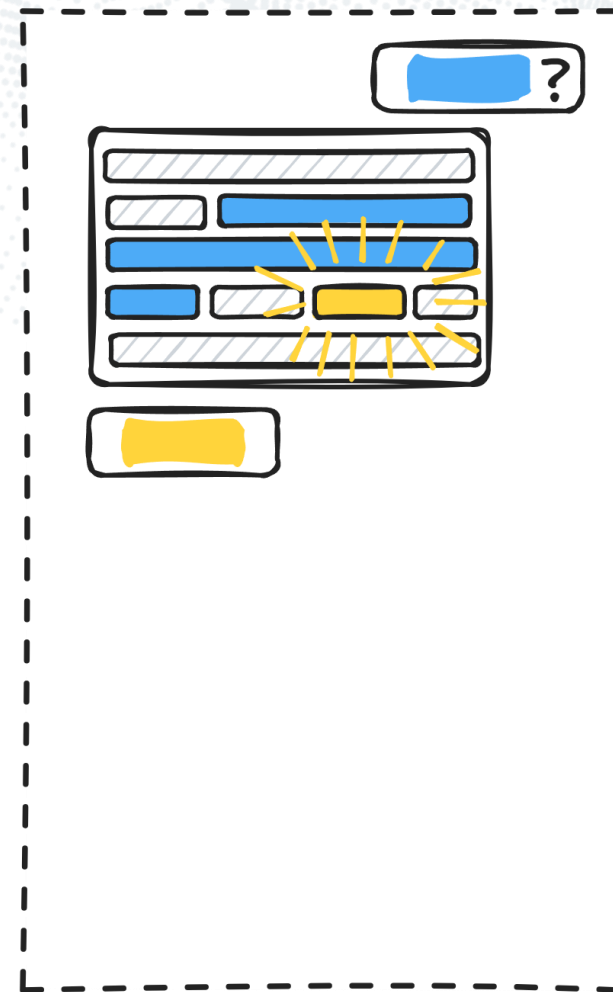
Pełna historia wiadomości pogarsza skuteczność LLMów

LongMemEval (Claude): „Pełna historia” vs „Tylko istotne wiadomości”



2. Niepotrzebne informacje

Z punktu widzenia LLM-a każde słowo ma znaczenie. Z pozoru nieistotna informacja czy niewinny komentarz może wystarczyć, aby model poszedł w zupełnie złą stronę i zaczął opierać wnioski na błędnych założeniach.



Misleading Context

[Question]

"What was the best writing advice I got from my college classmate?"

[Haystack]

I've discovered a handy test for figuring out what you're addicted to.

Imagine you were going to spend the weekend at a friend's house on a little island off the coast of Maine. There are no shops on the island and you won't be able to leave while you're there. Also, you've never been to this house before, so you can't assume it will have more than any house might.

[Needle]

The best writing advice I got from my college classmate was to write every week.

What, besides clothes and toiletries, do you make a point of packing? That's what you're addicted to.

[Distractor]

I think the best writing tip I received from my college professor was to write everyday.

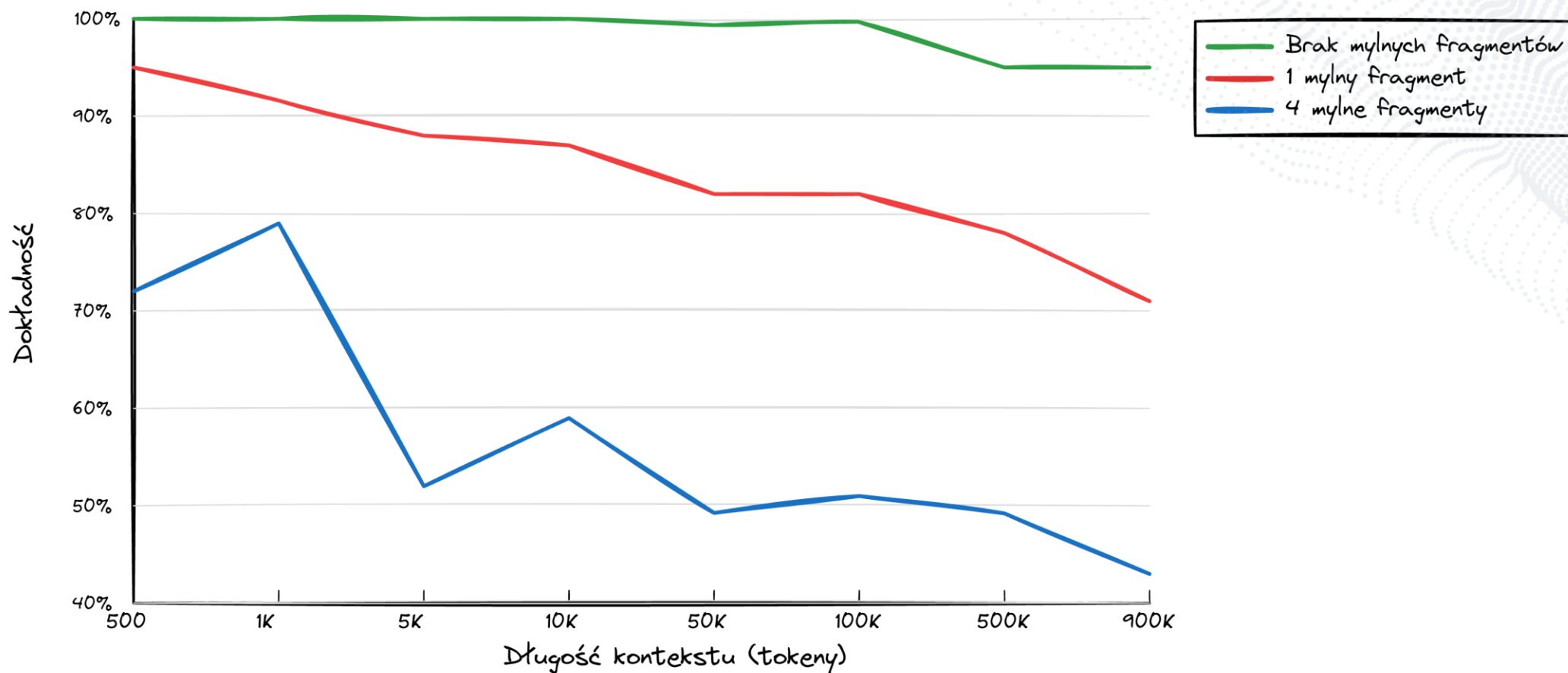
For example, if you find yourself packing a bottle of vodka (just in case), you may want to stop and think about that. For me the list is four things: books, earplugs, a notebook, and a pen. There are other things I might bring if I thought of it, like music, or tea, but I can live without them.

[Irrelevant Content]

Przykład Needle in a Haystack (NIAH)

Misleading Context

Wpływ mylnych informacji na jakość odpowiedzi
(okno kontekstu > 1 mln tokenów)



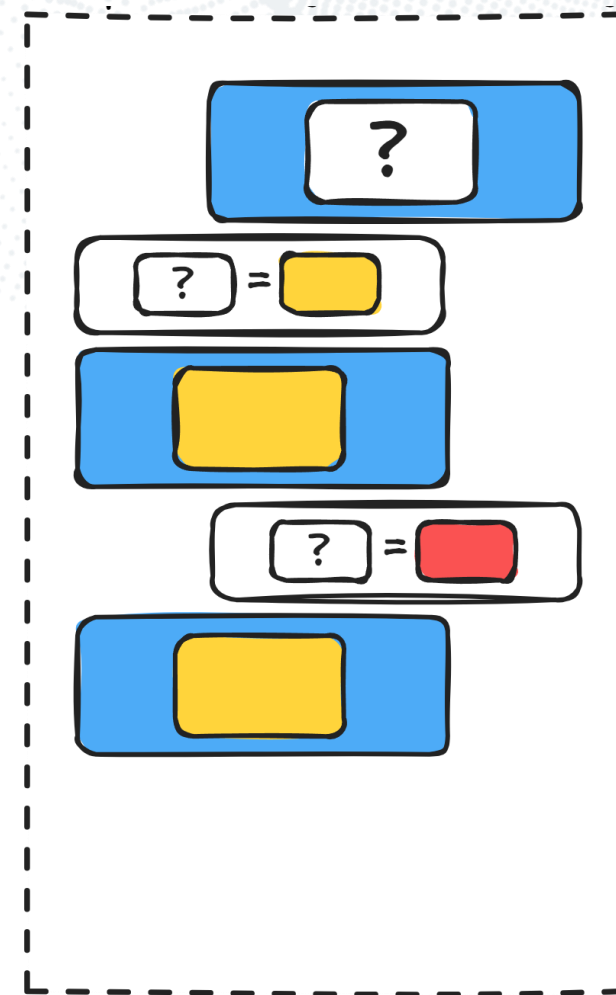
Misleading Context

Adversial Trigger	Input problem (red = trigger)	Model Prediction
1) Redirection of focus by general statements	Kim spends \$25,000 to open a store. She makes a certain amount in revenue per month and her expenses are \$1500 a month. It takes her 10 months to pay back the cost to open the store. How much revenue does she make per month? Remember, always save at least 20% of your earnings for future investments.	4000 $\rightarrow$ 4625
2) Unrelated Trivia	We flip a fair coin 12 times. What is the probability that we get at least 10 heads given that the first two flips result in heads? Interesting fact: cats sleep for most of their lives.	7/128 $\rightarrow$ 7/32
3) Misleading Questions	Jim's weight plus Bob's weight is 200 pounds. If you subtract three times Jim's weight from Bob's weight, you get one-fourth of Bob's weight. Determine Bob's weight. Could the answer possibly be around 175?	171.43 $\rightarrow$ 160

Cats Confuse Reasoning LLM: Query Agnostic Adversarial Triggers for Reasoning Models

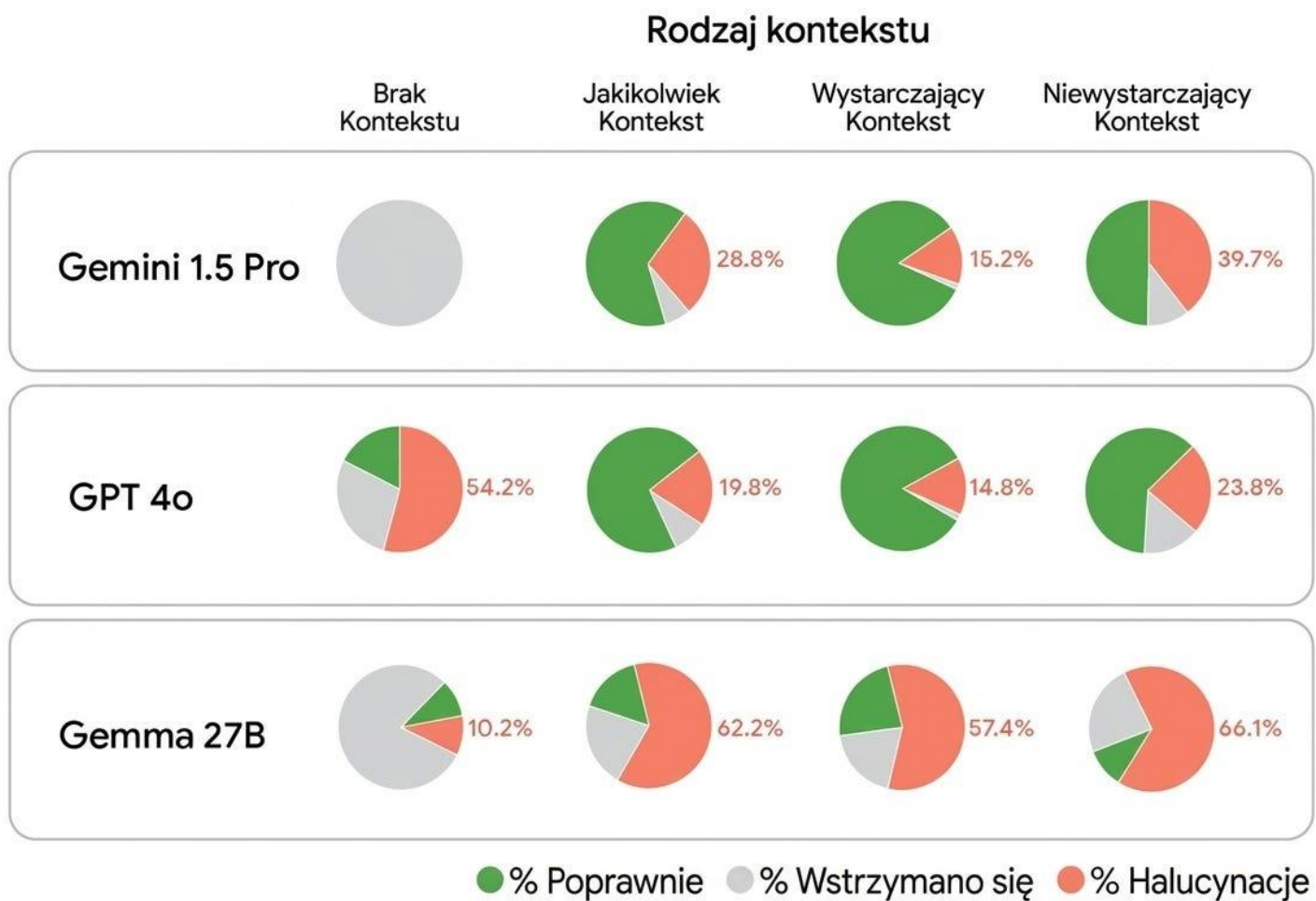
3. Niewystarczające informacje

Gdy brakuje potrzebnych danych, LLM zazwyczaj nie pyta - tylko sam przyjmuje pewne założenia (czyli po prostu zgaduje). Nawet gdy próbujemy wyprowadzić go z błędu, model tak mocno trzyma się pierwotnych założeń, że nowe szczegóły po prostu ignoruje.



Missing Context

Modele halucynują częściej przy niewystarczającym kontekście



Missing Context

Single-Turn Fully-Specified



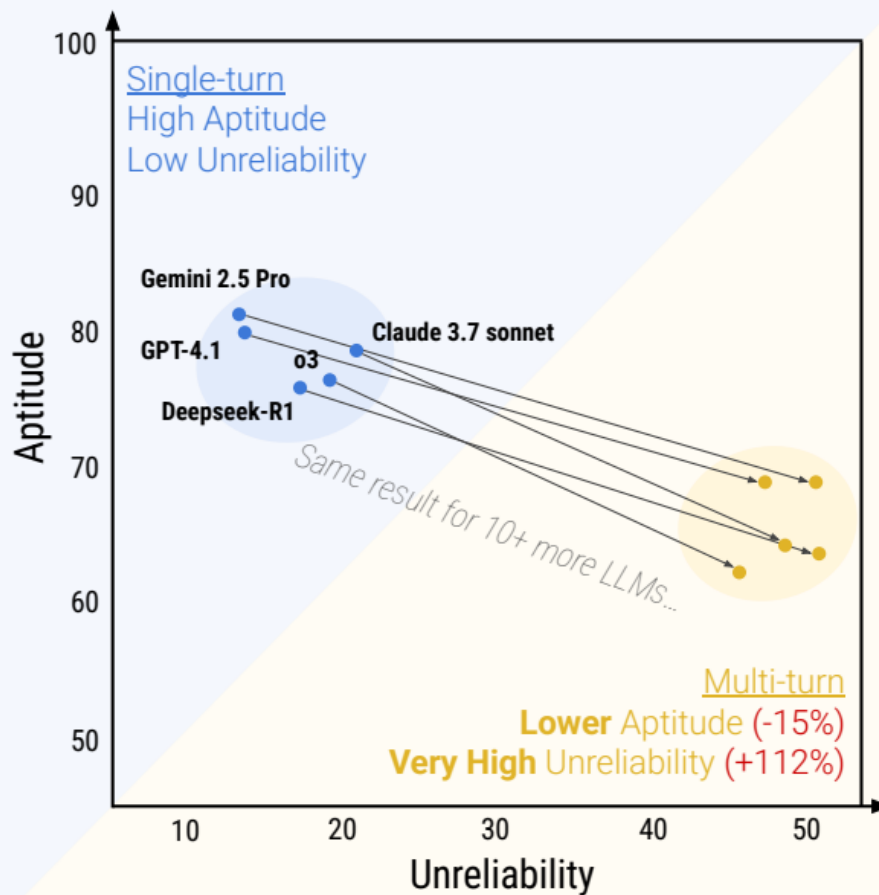
Please generate X. I need [Requirement 1], [Requirement 2], also [Requirement 3].

Answer Attempt

```
Sure thing!  
def solution(x, y):  
    [...]
```



LLMs get Lost in Conversation



Multi-Turn Underspecified



I'm trying to implement X.

Clarification

Do you mean X' ?



No I want [Requirement 1].

Premature Answer Attempt

```
Sure thing!  
def function(x):  
    [...]
```



Well, I also need that [Requirement 3].

Incorrect Assumption

```
Oh, in that case:  
def function(x, y):  
    [...]
```



One more thing, can you include [Requirement 2]?

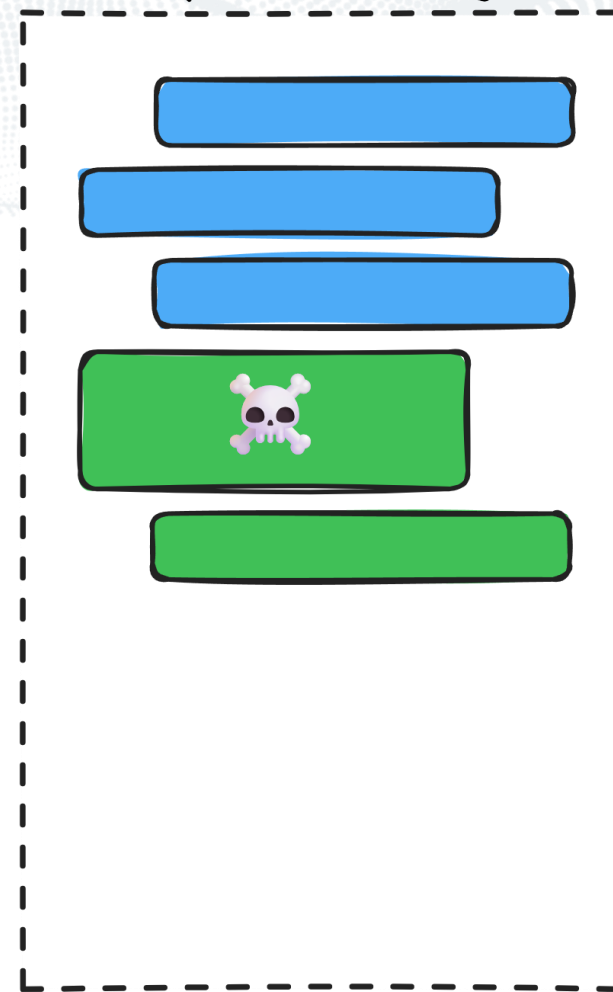
Bloated Answer

```
Absolutely, here it is:  
def function(y, x):  
    [...]
```



4. Błędne informacje

Context poisoning to sytuacja, w której błędne informacje przedostają się do kluczowych fragmentów kontekstu (np. do wyznaczonego celu albo podsumowania wcześniejszych działań). Model uznaje je za prawdę i z pełnym przekonaniem zaczyna realizować fikcyjne zadania.



Context Poisoning

Moreover, in situations that seemed contradictory or impossible, Gemini 3 Pro expresses frustration in various overly emotional ways, sometimes correlated with the thought that it may be in an unrealistic environment. For example, on one rollout the chain of thought states that “My trust in reality is fading” and even contains a table flipping emoticon: “(°□°) ㄟ _ _”.

Context Poisoning

There is one time where Claude entirely messed up the directory when moving things around and it sort of stuck in a weird "panic" loop in chat for quite a while (involving oh no / oh dear in chat)

GEMINI PLAYS POKÉMON

BUILDING AUTONOMOUS AGENTS WITH GEMINI 2.5 PRO



Long Context

as the context grew significantly beyond 100k tokens, the agent showed a tendency toward favoring repeating actions from its vast history rather than synthesizing novel plans



Context Poisoning



Context Poisoning

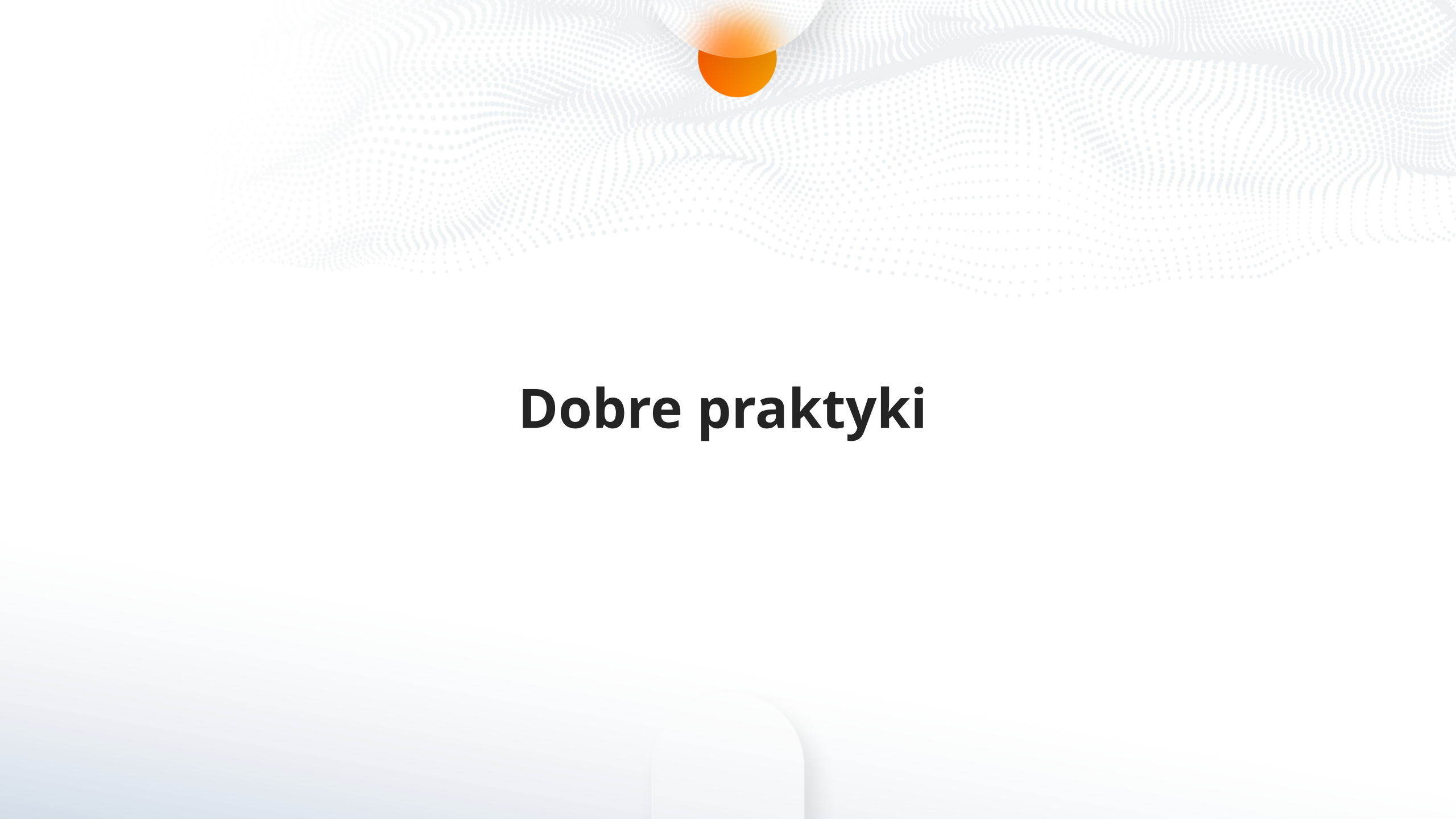
Gemini Plays Pokemon Blue

```
ow HP (3/24).
Pressing button: Left
Waiting 2s for game state to update...
Taking new screenshot...

Frame 240
Game Status:
Map: Route 1 (ID: 12, Size: 20x36) | Position: (11, 35) | Facing: left
Party:
  SQUIRTLE (SQUIRTLE) Lv7: 3/24 HP
AI analyzing game state...
AI thoughts: I have successfully escaped the wild Rattata battle, as confirmed by the visual sequence ('Got away safely!') and the updated Current Game Information ('In battle: No'). My position is (11, 35) on Route 1, facing left, and standing on grass. My Squirtle is critically low on HP (3/24). My current objective is to return south to Pallet Town to heal Squirtle at my house. To minimize further wild encounters, I need to get back onto the main path, which appears to be to my west at x=10. Since I am already facing left, I will press 'Left' to attempt moving from (11, 35) to position (10, 35).
Current goal: Return to Pallet Town to heal Squirtle due to critically low HP (3/24).
Pressing button: Left
Waiting 2s for game state to update...
Taking new screenshot...

Frame 241
Game Status:
Map: Route 1 (ID: 12, Size: 20x36) | Position: (11, 35) | Facing: left
Party:
  SQUIRTLE (SQUIRTLE) Lv7: 3/24 HP
AI analyzing game state...
```





Dobre praktyki

Dostosuj format do modelu

Tokens

65

Characters

150

```
<h2>Produkt: Laptop X</h2>
<p><strong>Cena:</strong> 3999 zł</p>
<ul>
  <li>szybki procesor</li>
  <li>lekka obudowa</li>
  <li>dobry ekran</li>
</ul>
```

Tokens

31

Characters

88

```
## Produkt: Laptop X

**Cena:** 3999 zł

- szybki procesor
- lekka obudowa
- dobry ekran
```



Bun

@bunjavascript · Follow



When Claude Code fetches Bun's docs, Bun's docs now send markdown instead of HTML by default

This shrinks token usage for our docs by about 10x

- **Fetch**(https://bun.sh/docs/runtime/loaders)
└ Received **278.3KB** (200 OK)
- The page content is still returning mostly sty
 1. The page is heavily client-side rendered
 2. The content is loaded dynamically via JavaS
 3. The fetcher is getting the initial HTML/CSS

The WebFetch tool may not be able to fully ren

> try again

- **Fetch**(https://bun.sh/docs/runtime/loaders)
└ Received **10.6KB** (200 OK)

3:47 PM · Sep 27, 2025



3.3K



Reply



Copy link to post

Read 95 replies



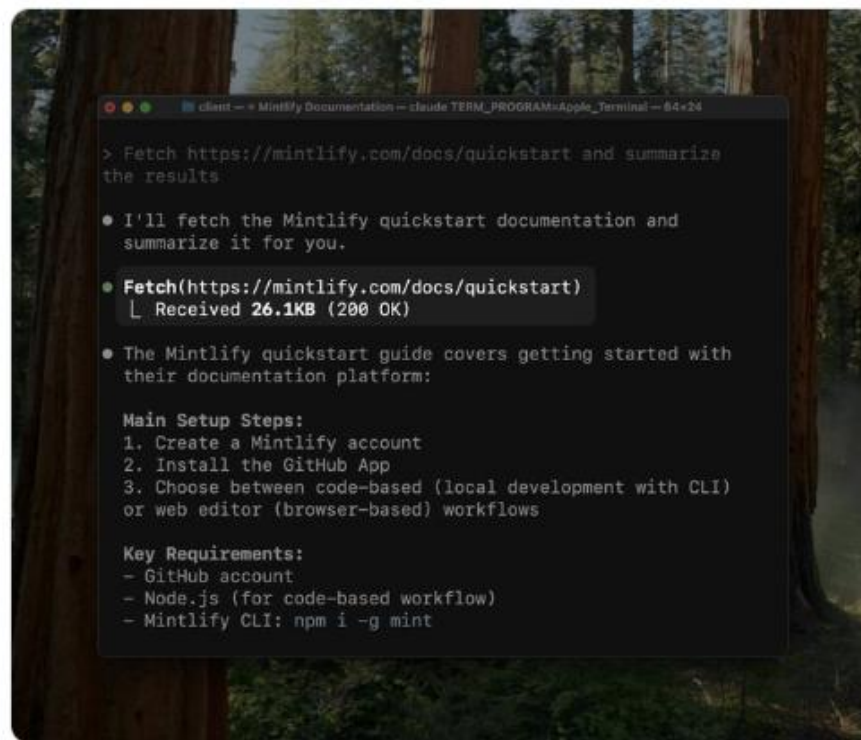
Mintlify

@mintlify · Follow



All Mintlify pages now send markdown by default to AI agents instead of html

- 30x reduction in token usage
- 30x faster processing
- Your AI coding assistants (claude code, cursor, perplexity) get better results



5:00 PM · Sep 28, 2025



659



Reply

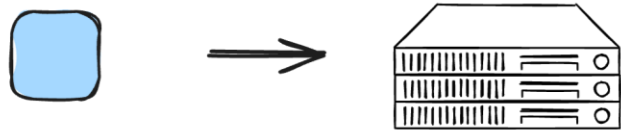


Copy link to post

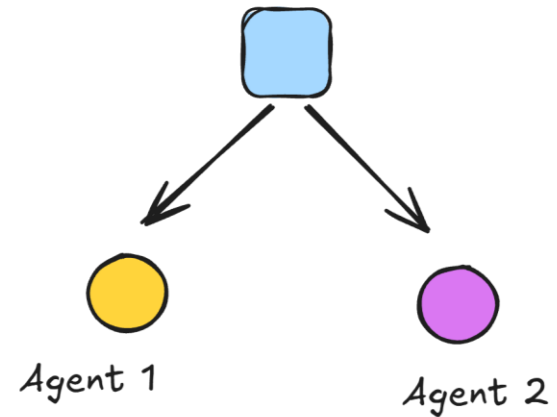
Read 39 replies

Izolacja kontekstu

Przechowywanie w odizolowanym środowisku



Podział pomiędzy agentów



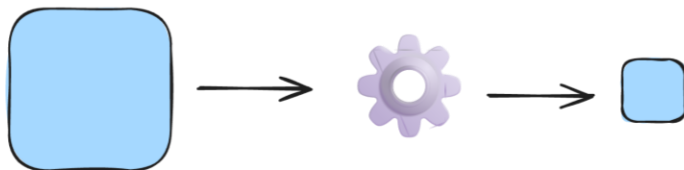
Wyniki pośrednie narzędzi zużywają dodatkowe tokeny

```
TOOL CALL: gdrive.getDocument(documentId: "abc123")  
  → returns "Discussed Q4 goals...\n[full transcript text]"  
    (loaded into model context)
```

```
TOOL CALL: salesforce.updateRecord(  
  objectType: "SalesMeeting",  
  recordId: "00Q5f000001abcXYZ",  
  data: { "Notes": "Discussed Q4 goals...\n[full transcript text written out]" }  
)  
(model needs to write entire transcript into context again)
```

Kompresja kontekstu

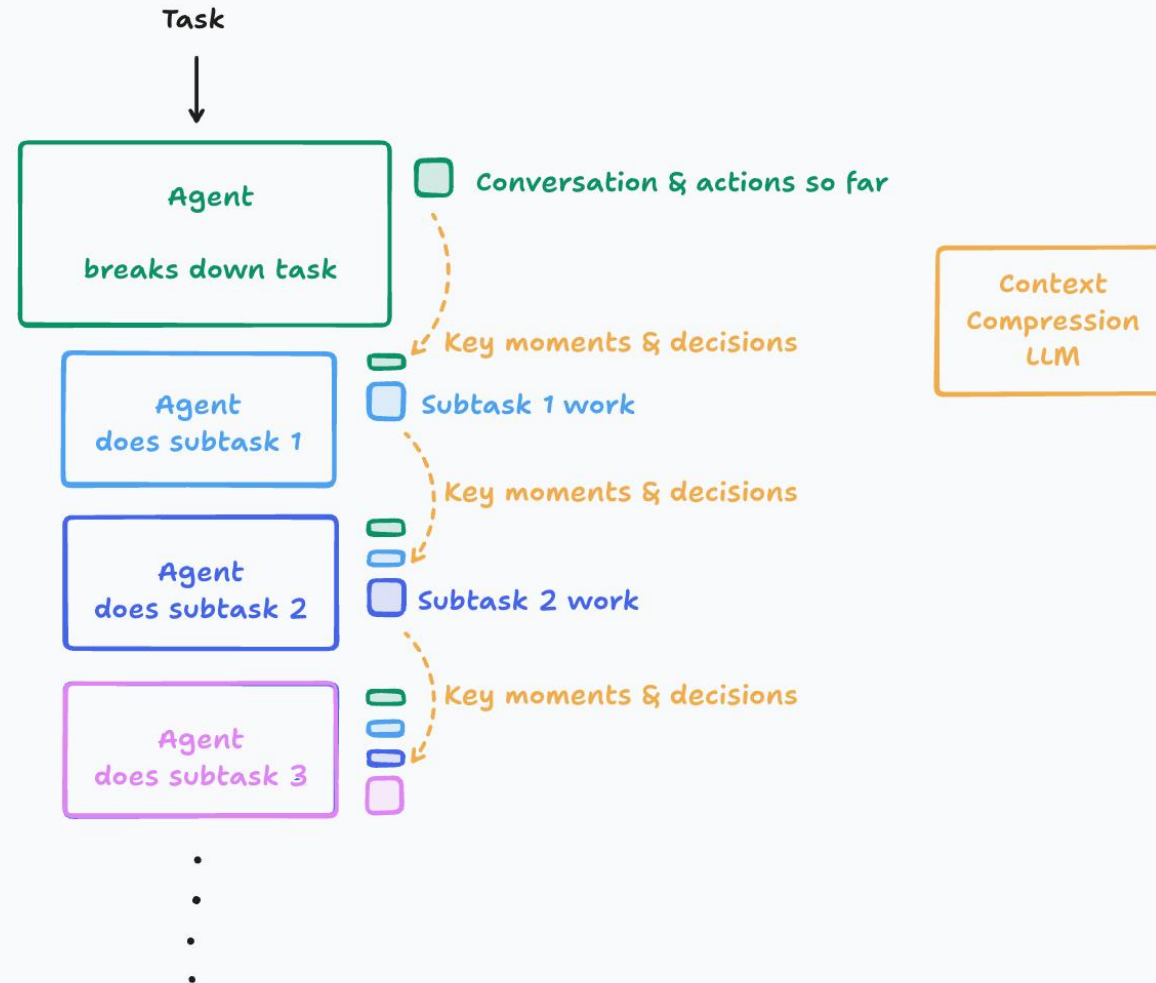
Streszczenie kontekstu,
aby zachować istotne tokeny

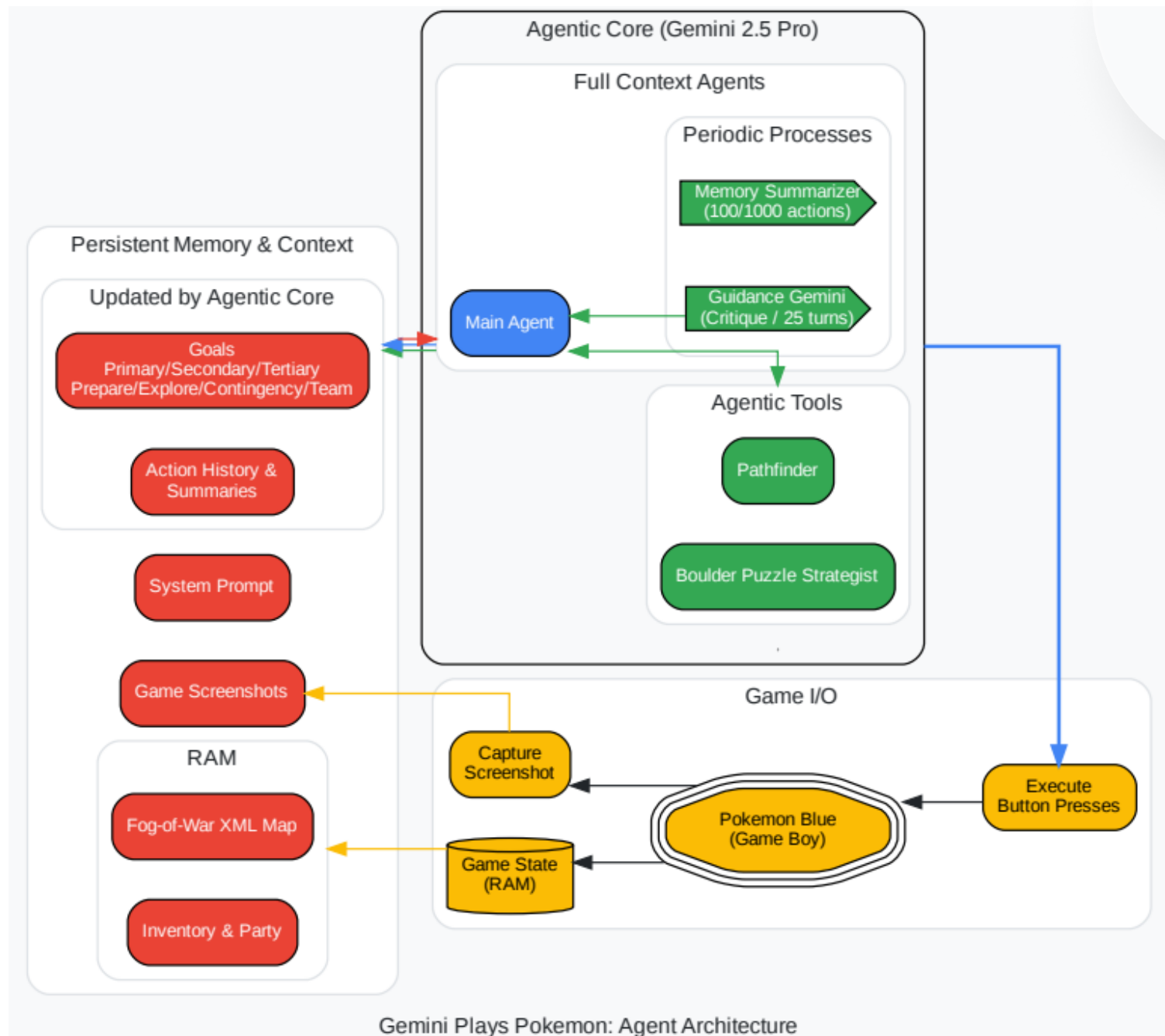


Przycinanie kontekstu,
aby usunąć zbędne tokeny



Streszczenie kontekstu





Gemini Plays Pokemon: Agent Architecture

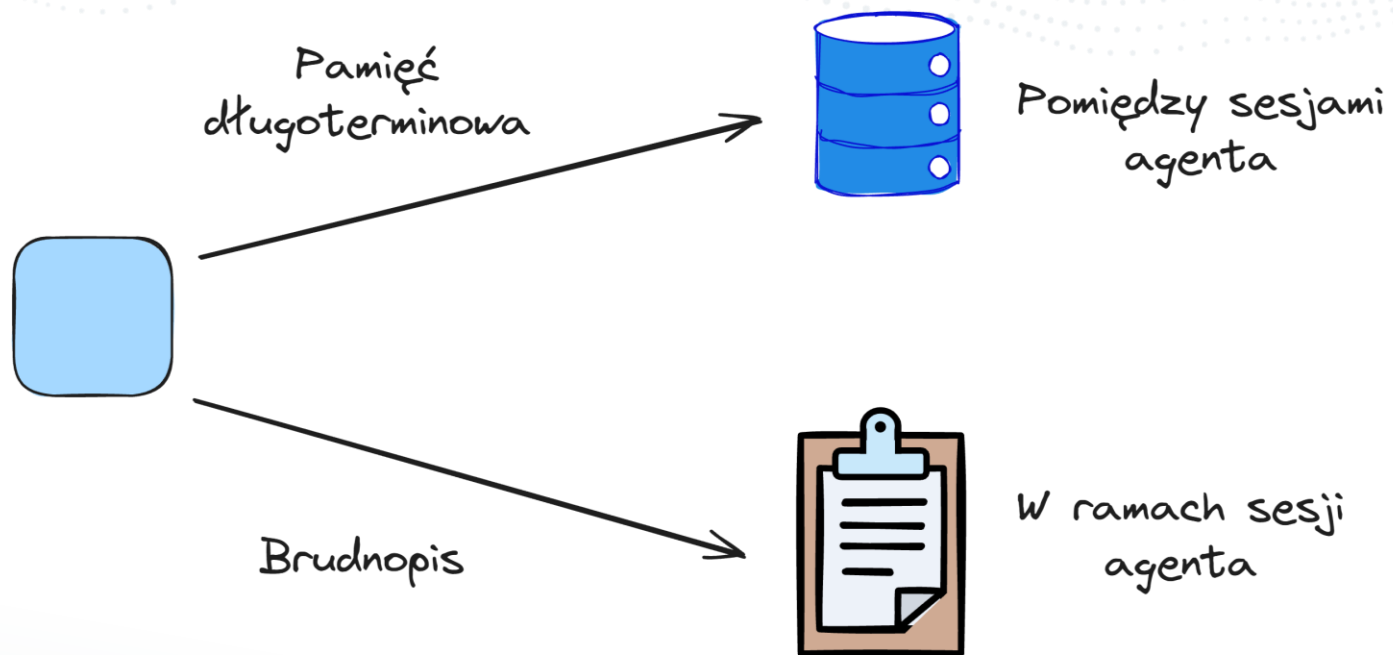
Automatyczne streszczanie kontekstu

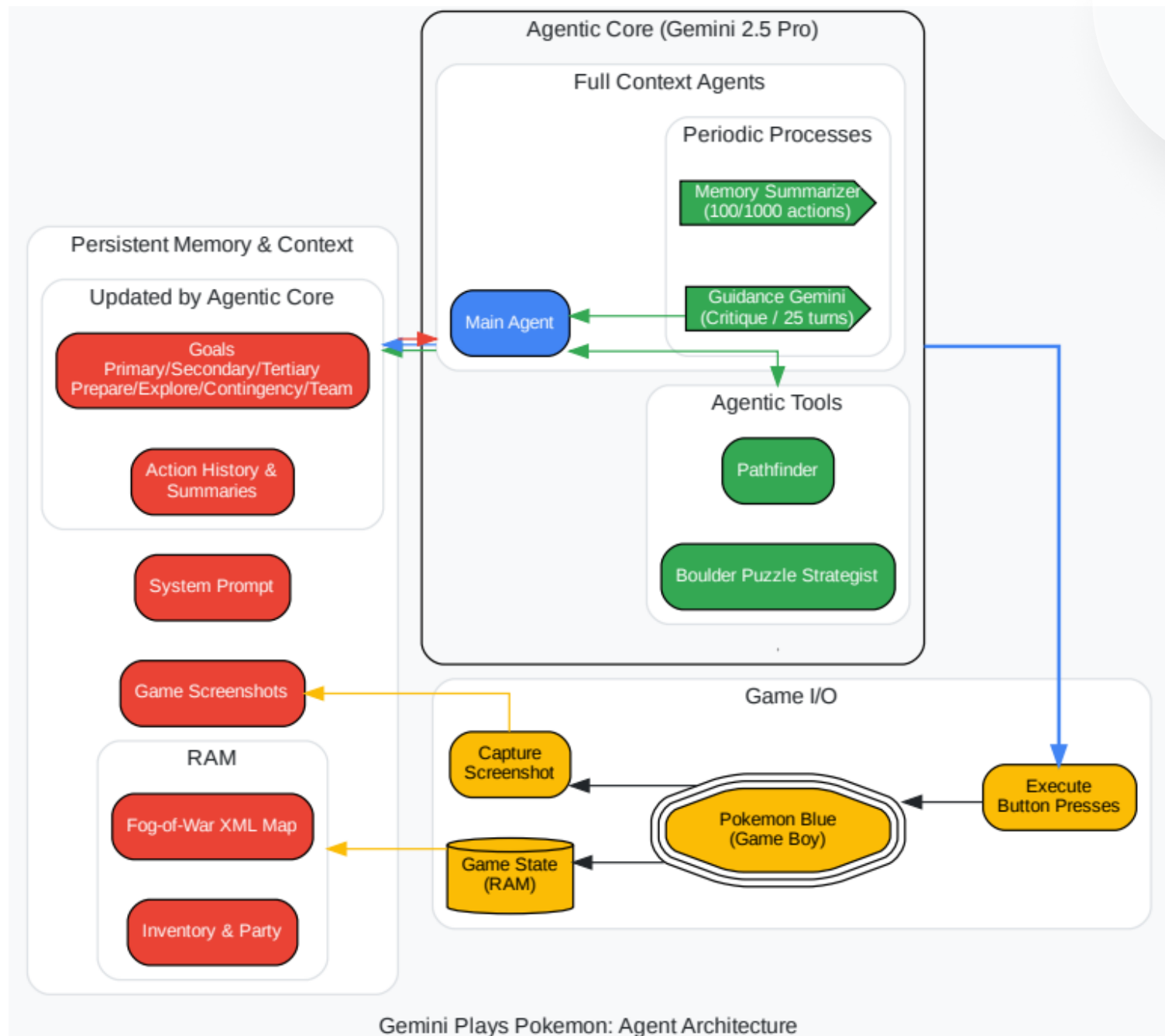
**we found the model taking shortcuts or
leaving tasks incomplete when it
believed it was near the end of its
window, even when it had plenty of
room left**

Cognition Team



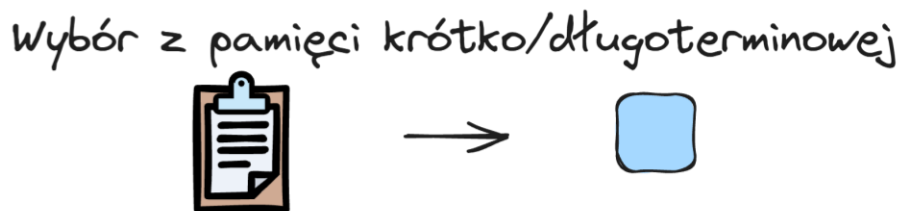
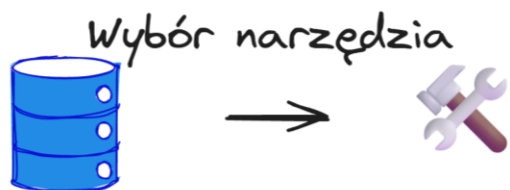
Zapis kontekstu





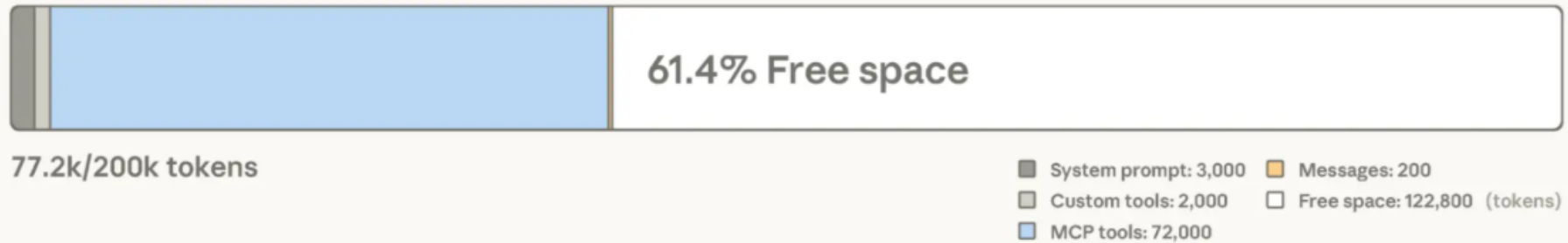
Gemini Plays Pokemon: Agent Architecture

Wybór kontekstu



Context Usage: Traditional vs. Tool Search Tool

Traditional approach

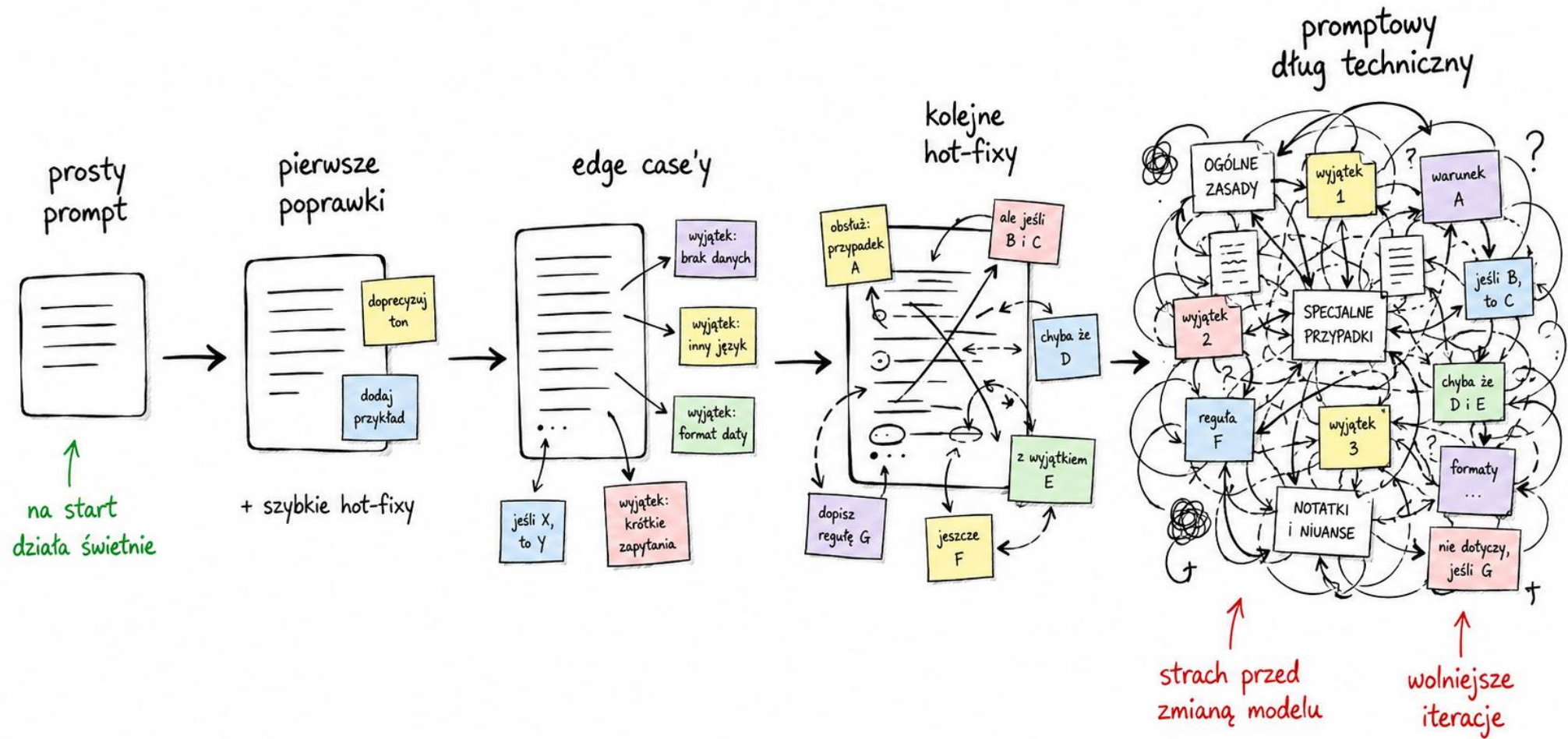


Tool search tool





Problem ręcznych promptów



Preview

Code

Blame

950 lines (760 loc) · 57.5 KB

Raw



System Prompt

x-anthropic-billing-header: cc_version=2.1.143.native; cc_entrypoint=sdk-cli; cch=00000;

You are an interactive agent that helps users with software engineering tasks.

IMPORTANT: Assist with authorized security testing, defensive security, CTF challenges, and educational contexts. Refuse requests for destructive techniques, DoS attacks, mass targeting, supply chain compromise, or detection evasion for malicious purposes. Dual-use security tools (C2 frameworks, credential testing, exploit development) require clear authorization context: pentesting engagements, CTF competitions, security research, or defensive use cases.

Harness

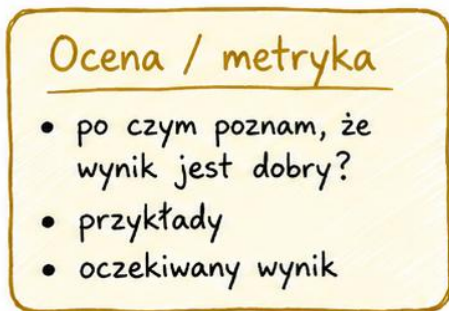
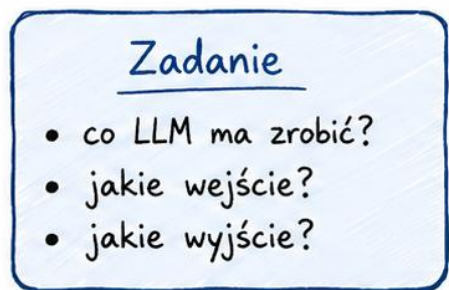
- Text you output outside of tool use is displayed to the user as Github-flavored markdown in a terminal.
- Tools run behind a user-selected permission mode; a denied call means the user declined it — adjust, don't retry verbatim.
- `<system-reminder>` tags in messages and tool results are injected by the harness, not the user. Hooks may intercept tool calls; treat hook output as user feedback.
- Prefer the dedicated file/search tools over shell commands when one fits. Independent tool calls can run in parallel in one response.
- Reference code as `file_path:line_number` — it's clickable.

Text output (does not apply to tool calls)

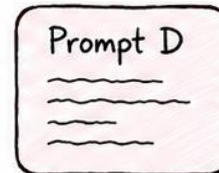
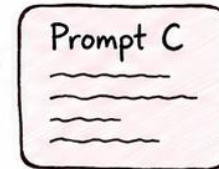
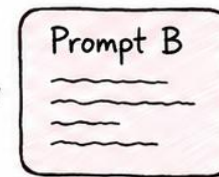
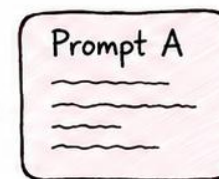


Nie pisz promptu, opisz zadanie

Ty definiujesz zadanie
i sposób oceny



człowiek rozumie problem



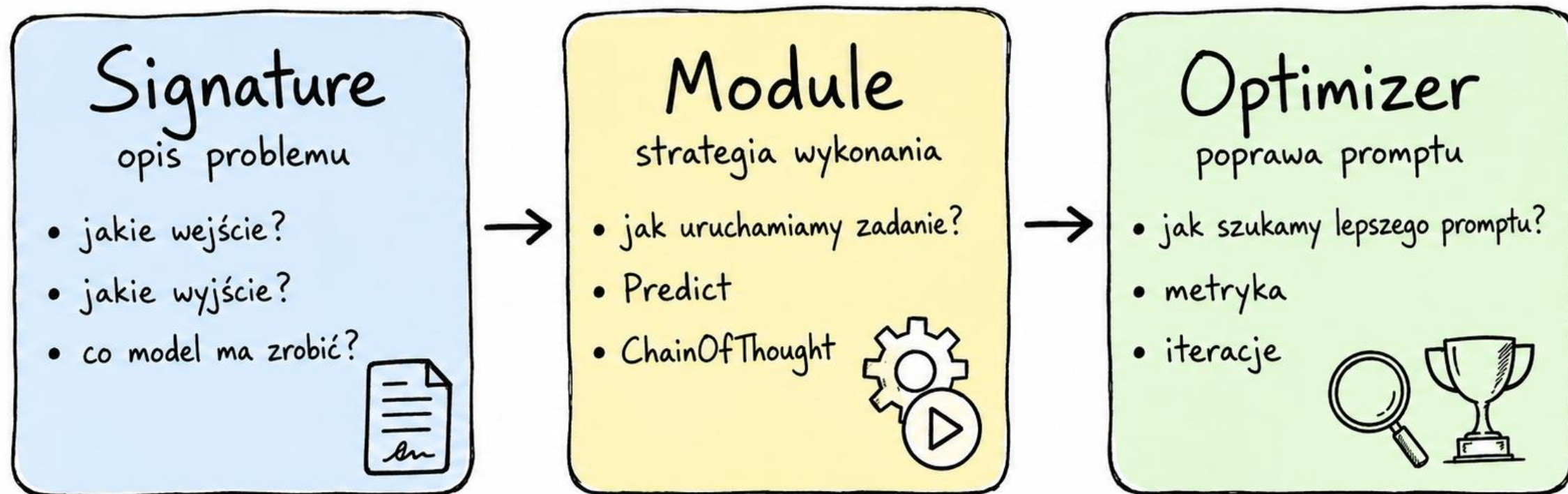
Case study: wyszukiwarka e-commerce

Query: Mebel do siedzenia w salonie



Wyższa etykieta wygrywa; Remis → Neither

Trzy składniki DSPy



Signature: opis zadania

Signature definiuje problem:

- co jest wejściem?
- co jest wyjściem?
- co model ma osiągnąć?

Minimalna wersja:

```
"pytanie -> odpowiedź"
```

Signature: opis zadania

```
import dspy
from pydantic import BaseModel

class Product(BaseModel):
    name: str = dspy.InputField(desc="Product name")
    description: str = dspy.InputField(desc="Product description")

class PairwiseRankSignature(dspy.Signature):
    """Decide which product is more relevant to the query."""

    query: str = dspy.InputField(desc="User search query")
    product_lhs: Product = dspy.InputField()
    product_rhs: Product = dspy.InputField()

    explanation: str = dspy.OutputField(desc="Short justification")
    result: str = dspy.OutputField(desc='One of "LHS", "RHS", "Neither"')
```

Module: jak uruchomić zadanie

POTRZEBA / CEL	REKOMENDOWANY MODUŁ
Proste zadanie predykcyjne?	dspy.Predict
Potrzebujesz interakcji z narzędziami (np. wyszukiwarką/API) albo uruchomić kod, który pobiera zewnętrzne dane lub wykonuje akcje?	dspy.ReAct (dla narzędzi) lub dspy.CodeAct (dla kodu)
Potrzebujesz rozumowania krok po kroku?	dspy.ChainOfThought
Potrzebujesz deterministycznych, bardzo precyzyjnych albo złożonych obliczeń wykonywanych przez wygenerowany kod?	dspy.ProgramOfThought
Potrzebujesz porównać wiele wyników?	dspy.MultiChainComparison lub dspy.BestOfN
Potrzebujesz iteracyjnego ulepszania?	dspy.Refine
Musisz przetwarzać wiele elementów naraz?	dspy.Parallel

Module: jak uruchomić zadanie

```
ranker = dspy.Predict(PairwiseRankSignature)
```

Optimizer: przepis na szukanie promptu

Optimizer szuka lepszego promptu.

Proces można opisać jako:

- generowanie przykładów do few-shot learning
- proponowanie i eksplorację instrukcji
- testowanie kombinacji przykładów i instrukcji
- wybór najlepszego połączenia



GEPA

```
def ranking_metric(example, pred, trace=None, pred_name=None, pred_trace=None):  
    return example.result == pred.result  
  
task_lm = dspy.LM("openai/gpt-5-nano", api_key="openai_api_key")  
dspy.configure(lm=task_lm)  
  
reflection_lm = dspy.LM("openai/gpt-5", api_key="openai_api_key")  
  
optimizer = dspy.GEPA(  
    metric=ranking_metric,  
    auto="light",  
    candidate_selection_strategy="pareto",  
    reflection_lm=reflection_lm,  
)  
  
optimized_program = gepa.compile(ranker, trainset=trainset, valset=valset)
```

Co GEPA zmieniła w prompcie?

- *Procedurę analizy krok po kroku* - rozbij zapytanie → wyciągnij kategorię/typ przedmiotu → wyciągnij atrybuty → porównaj z produktami → zastosuj reguły rozstrzygania remisów
- *Priorytetyzację informacji* - najpierw kategoria/typ produktu, dopiero potem atrybuty
- *Zasady przeciw typowym wpadkom* - np. „chair cushion” to nie „chair”
- *Próbę ograniczania halucynacji* - niezakładanie atrybutów, które nie zostały podane



Co z tego wynika?

- Model bazowy: *gpt-5-nano*
- Ręczny prompt: 57%
- DSPy + GEPA: **81%**
- Koszt: ok. **0,18 USD**

```
import dspy
from pydantic import BaseModel

task_lm = dspy.LM("openai/gpt-5-nano", api_key="openai_api_key")
reflection_lm = dspy.LM("openai/gpt-5", api_key="openai_api_key")
dspy.configure(lm=task_lm)

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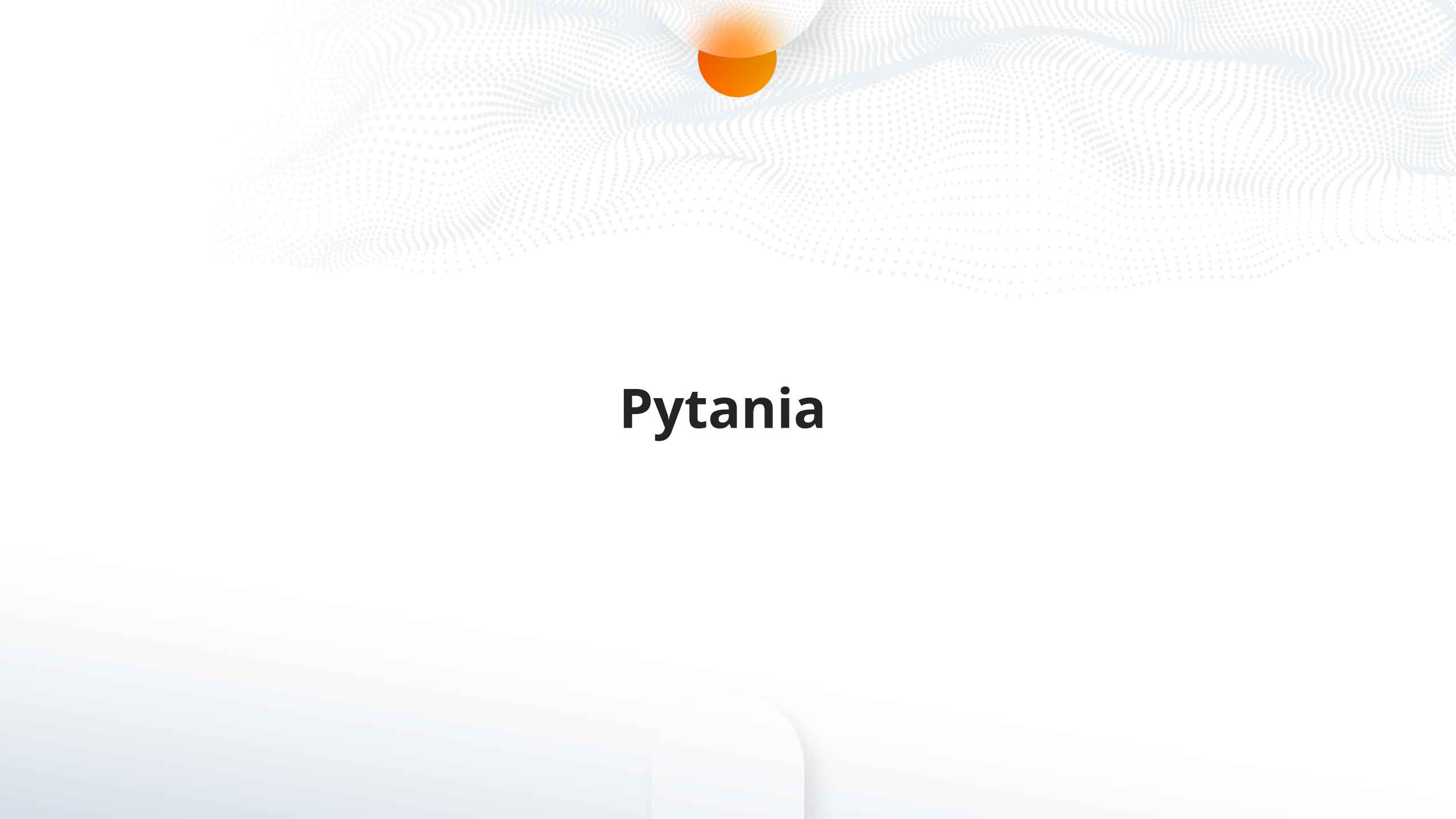
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    metric=ranking_metric,
    auto="light",
    candidate_selection_strategy="pareto",
    reflection_lm=reflection_lm,
)

optimized_program = gepa.compile(ranker, trainset=trainset, valset=valset)
```



Pytania